

MACRO L3 - 2014-2015

Lezione 5

F. PORTIER

Session 4

These notes are those I wrote during class. They are posted "for the record".

Session 4

Nov 13, 2014

Utility maximisation

$$\max_{n_t, m_t} m_t + \alpha \log n_t$$

$$\text{s.t. } 1 \times m_t + p_t n_t = w_t$$

$$\Leftrightarrow \max_{n_t} w_t - p_t n_t + \alpha \log n_t$$

$$\underline{\text{FOC}} \quad -p_t + \frac{\alpha}{n_t} = 0 \Leftrightarrow n_t = \frac{\alpha}{p_t}$$

optimal choice for m_t ?

Budget constraint:

$$m_t = w_t - p_t n_t$$

$$= w_t - p_t \times \frac{\gamma}{p_t}$$

$$m_t = w_t - \gamma$$

$$A_{t+1} - A_t = y_t^A$$

$$g_t^A = \frac{A_{t+1} - A_t}{A_t} = \frac{y_t^A}{A_t} = \frac{A_t^\varepsilon L_t^{A^d} 1^{1-d}}{A_t}$$

$$g_t^A = A_t^{\varepsilon-1} L_t^{A^d}$$

$$g_t^n = \frac{n_{t+1} - n_t}{n_t} = \frac{y_t^m}{n_t}$$

$$g_t^n = \frac{n_t^{\varepsilon} L_t^n}{n_t} = n_t^{\varepsilon-1} L_t^n$$

Period t

State variables L_t, A_t, Π_t

They cannot be
changed in t

Adults choose

- where they work
- how much children and clothes they "buy" = food

many good

They choose given prices (P_t, w_t)

then we look for (p_t, w_t)

s.t

the M market
the food market
the labor market } clear

• This determines

$$\begin{aligned} n_t &\rightarrow L_{t+1} \\ y_t^A &\rightarrow A_{t+1} \\ y_t^N &\rightarrow n_{t+1} \end{aligned}$$